2025 1st IEEE Uttar Pradesh Section Women in Engineering International Conference on Electrical Electronics and Computer Engineering (UPWIECON)

Individual Load Monitoring using Smart Energy Meter: An IoT-based Approach

Abhijeet Singh
ASH (Electronics) Department
REC Banda
(AKTU Lucknow)
Banda, India
ORCID: 0009-0007-2877-1041

Pushpendra Singh
Electrical Engg. Department
REC Banda
(AKTU Lucknow)
Banda, India
erpsingh@rediffmail.com

Neelam Srivastava
Electronics & Communication Engineering Department
I.E.T., Lucknow
Lucknow, India
neelam02Srivastava@gmail.com

Anurag Chauhan
Electrical Engg. Department
REC Banda
(AKTU Lucknow)
Banda, India
anurag@recbanda.ac.in

Vipin Agrahari
Electrical Engg. Department
REC Banda
(AKTU Lucknow)
Banda, India
vipinagrahari2014@gmail.com

Shweta Sonkar
Electrical Engg. Department
REC Banda
(AKTU Lucknow)
Banda, India
shwetasonkar15@gmail.com

Sakshi Tripathi
Electrical Engg. Department
REC Banda
(AKTU Lucknow)
Banda, India
tripathisakshi1905@gmail.com

Shivani Gautam
Electrical Engg. Department
REC Banda
(AKTU Lucknow)
Banda, India
shivani.gautam2001734@gmail.com

***Abstract*— This paper presents a comprehensive IoT-based smart energy monitoring system designed for individual load monitoring and intelligent control in residential and small commercial applications. The proposed system integrates multiple PZEM004T sensors with microcontroller units to enable real-time monitoring of critical electrical parameters, including voltage, current, power factor, and energy consumption at the individual appliance level. The multi-tier architecture incorporates cloud connectivity through ThingSpeak for data analytics and Arduino Cloud for remote control, providing consumers with granular insights into their energy consumption patterns. The system demonstrates the capability for remote monitoring and control through both web interfaces and mobile applications, thereby promoting energy efficiency awareness and enabling sustainable consumption practices through informed decision-making.**

Keywords*— *Smart energy meter, IoT, Individual load monitoring, PZEM004T, Arduino UNO, ThingSpeak, Arduino Cloud.

I. Introduction

The exponential growth in global electricity demand, coupled with increasing environmental concerns and the need for efficient energy management, has catalyzed the development of intelligent power monitoring systems integrated with smart grid infrastructure. Traditional energy metering systems, although functional for billing purposes, offer limited visibility into individual appliance consumption patterns, resulting in a significant gap in consumer awareness and energy optimization opportunities [1-5].

The evolution of electricity metering technology has progressed through distinct phases: electromechanical meters with rotating discs, electronic meters with digital displays, and contemporary smart metering systems integrated with Advanced Metering Infrastructure (AMI). While electromechanical meters, despite their widespread deployment in residential sectors, are constrained by measurement accuracy limitations and can only quantify active power components, electronic meters introduced through Automated Meter Reading (AMR) systems have improved measurement precision and reduced operational overhead. The latest generation of Smart Metering Systems (SMS) provides comprehensive bidirectional communication capabilities and enhanced monitoring features.

However, current digital energy meters typically display aggregate power consumption, leaving consumers uninformed about individual appliance energy usage patterns and power quality characteristics. This limitation becomes particularly significant in the context of modern residential and commercial environments where diverse electrical loads with varying power factor characteristics coexist. Poor power factor not only increases energy costs but also contributes to system inefficiencies and potential equipment damage.

The proposed research addresses these limitations by developing an IoT-enabled energy monitoring and control system capable of device-level power consumption measurement, power factor monitoring, and intelligent load management. The system's distributed sensor architecture enables comprehensive monitoring of individual loads while maintaining overall system visibility, thereby empowering consumers with actionable insights for energy optimization [6].

Recent developments in smart metering technology have focused on integrating IoT capabilities with traditional energy monitoring systems. Karpagam et al. [7] developed a smart energy meter emphasizing theft prevention and cloud-based consumption monitoring, demonstrating the importance of security in smart metering applications. Mondal et al. [8] implemented a system enabling real-time

979-8-3315-6628-9/25/$31.00 ©2025 IEEE

Authorized licensed use limited to: VTU Consortium. Downloaded on March 26,2026 at 15:28:08 UTC from IEEE Xplore. Restrictions apply.

2025 1st IEEE Uttar Pradesh Section Women in Engineering International Conference on Electrical Electronics and Computer Engineering (UPWIECON) | 979-8-3315-6628-9/25/$31.00 ©2025 IEEE | DOI: 10.1109/UPWIECON67212.2025.11390313

power consumption monitoring with remote load control capabilities, establishing the foundation for intelligent energy management.
Sebastian and Deepa [9] proposed an IoT-based smart meter incorporating integrated fault detection and theft prevention features, utilizing current sensors for comprehensive monitoring of electrical parameters. Their work highlighted the importance of system reliability and security in smart metering deployments. Selvam et al. [10] designed a cost-effective Smart Energy Management System for evaluating daily power usage patterns, demonstrating the economic viability of IoT-based solutions.

In terms of communication infrastructure development, Orlando et al. [11] developed a three-phase smart meter following IoT design principles, capable of executing multiple algorithms for smart grid management applications. This work established the scalability requirements for industrial and commercial applications. Saleem et al. [12] proposed a comprehensive Smart Energy Management System (SEMS) incorporating smart meters with IoT middleware for efficient data analysis and management, emphasizing the importance of data processing capabilities.

Addressing user interface and data visualization requirements, Sheeba et al. [13] developed a system for converting conventional static energy meters into smart meters with graphical user interface capabilities. Souza et al. [14] implemented a low-cost IoT-based meter providing online communication functionality, demonstrating the accessibility of smart metering technology.

Security considerations have been addressed by Hseiki et al. [15], who proposed a multi-level security architecture ensuring hardware, communication, and data security for smart energy meters. Their work specifically addressed DDoS attacks, data integrity issues, and energy theft concerns, establishing security as a fundamental requirement for smart metering systems.

The literature reveals a clear trend toward integrating IoT capabilities with energy monitoring systems, focusing on real-time monitoring, remote control, and security features. However, there remains a gap in systems that provide comprehensive individual load monitoring combined with power factor measurement and control capabilities in a cost-effective package. The proposed system addresses these gaps by providing a multi-sensor architecture that enables individual load monitoring while maintaining system-level visibility, thereby contributing to the advancement of intelligent energy management systems.

II. Problem Definition

Contemporary residential and commercial environments are characterized by an increasing number of electrical appliances with diverse power consumption patterns and varying power factor characteristics. The absence of individual load monitoring capabilities in existing energy metering systems creates several critical challenges:

1. Limited visibility into individual appliance energy consumption patterns.
2. Inability to identify power factor issues at the device level.
3. Lack of remote-control capabilities for load management.
4. Insufficient data for energy optimization decision-making.
5. Poor power quality awareness leading to increased energy costs.

The primary objective of this research is to develop and validate an IoT-based smart energy monitoring system that addresses the identified limitations through:

1. **Individual Load Monitoring**: Implementation of device-level monitoring for voltage, current, power factor, and energy consumption.
2. **Power Factor Analysis**: Real-time measurement and analysis of power factor characteristics for individual loads and overall system.
3. **Remote Control Capabilities**: Integration of cloud-based control systems for load management.
4. **Data Visualization**: Development of user-friendly interfaces for energy consumption analysis.
5. **System Integration**: Seamless integration of hardware components with cloud platforms for comprehensive energy management.

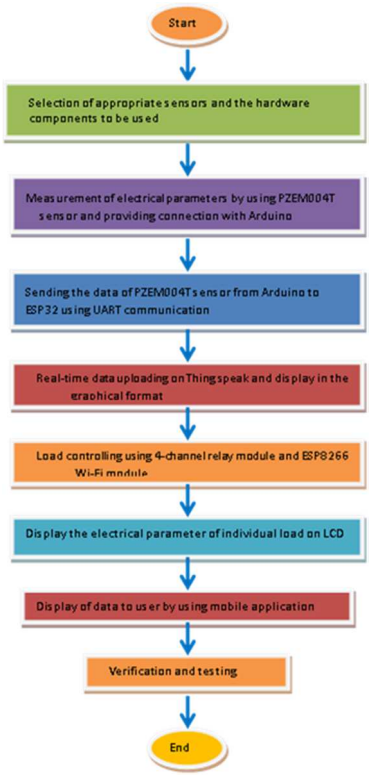

Figure 1. Flow chart of the proposed methodology

Authorized licensed use limited to: VTU Consortium. Downloaded on March 26,2026 at 15:28:08 UTC from IEEE Xplore. Restrictions apply.

Fig. 1, shows the flow chart of proposed methodology. Fig. 2, shows the block diagram of individual load monitoring using smart energy meter.

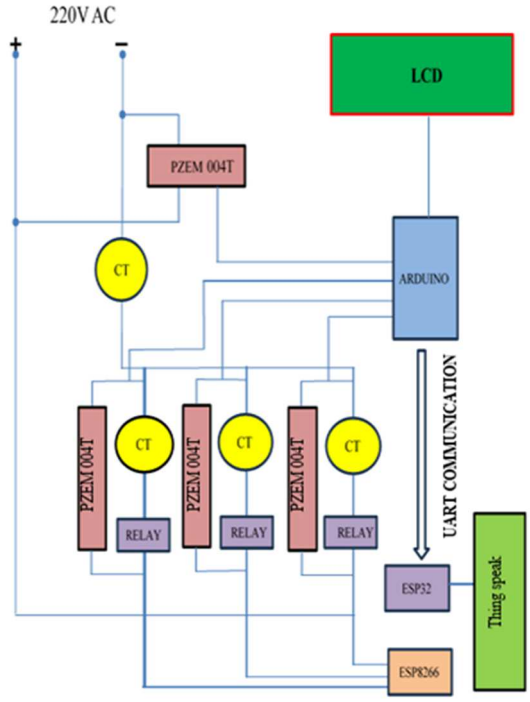

Figure 2: Block diagram of individual load monitoring using smart energy meter

III. System Architecture

The proposed system consists of the following key components:

A. Hardware Architecture

- Arduino UNO microcontroller: Primary system controller responsible for sensor data acquisition, processing, and communication coordination.
- ESP32S Module: Dedicated power factor monitoring and analysis unit with advanced processing capabilities
- ESP8266 Module: Wireless communication interface for load control and cloud connectivity
- PZEM004T 100A Sensors: Multi-parameter electrical measurement sensors capable of measuring voltage (80-260V), current (0-100A), power, energy, frequency, and power factor
- Distributed Sensor Configuration: Individual sensors dedicated to specific loads plus system-level monitoring
- 4-Channel Relay Module: Solid-state switching system for individual load control
- 20×4 LCD Display with I2C Interface: Local display system for real-time parameter visualization
- Communication Infrastructure: UART-based inter-module communication

B. Software Architecture

The software architecture implements a layered approach for data processing and system management:

1. *Data Acquisition Layer*
   - Sensor Interface Drivers: Low-level drivers for PZEM004T sensor communication.
   - Data Validation: Real-time data validation and error detection algorithms.
   - Parameter Calculation: Advanced algorithms for power factor analysis and energy consumption calculation.
2. *Communication Layer*
   - UART Protocol Implementation: Inter-module communication protocols.
   - WiFi Communication: Wireless connectivity for cloud integration.
   - Data Formatting: Standardized data formatting for cloud transmission.
3. *Cloud Integration Layer*
   - ThingSpeak Integration: Data visualization and analytics platform.
   - Arduino Cloud Integration: Remote control and monitoring platform.
   - Mobile Application Interface: User-friendly mobile access

IV. Methodology

A. System Implementation Approach

The system implementation follows a modular approach that enables scalable deployment and maintenance:

1. ***Data Acquisition and Processing***: The data acquisition subsystem implements a multi-sensor approach where individual PZEM004T sensors monitor specific loads while an additional sensor provides system-level measurements. This configuration enables:
   - Load-specific monitoring of electrical parameters
   - Power factor analysis at individual appliance level
   - System-level validation through aggregate measurements
   - Real-time data processing for immediate feedback.
2. ***Communication and Control***: The communication subsystem implements a hierarchical architecture:

Authorized licensed use limited to: VTU Consortium. Downloaded on March 26,2026 at 15:28:08 UTC from IEEE Xplore. Restrictions apply.

- Local Communication: UART-based communication between Arduino UNO and ESP modules
- Cloud Communication: WiFi-based connectivity to ThingSpeak and Arduino Cloud platforms
- Control Implementation: Relay-based switching for individual load control

3. ***Data Management and Visualization***: The data management system provides comprehensive analytics capabilities:

- Real-time Monitoring: Continuous parameter tracking and display
- Historical Analysis: Time-series data storage and analysis
- Threshold Management: Configurable alerts for power factor and consumption limits
- User Interface: Multi-platform access through web and mobile interfaces

B. Power Factor Analysis Methodology

The power factor analysis implementation addresses the critical need for power quality monitoring:

1. ***Individual Load Analysis***: Each PZEM004T sensor provides real-time power factor measurements for individual loads, enabling:

- Load characterization based on power factor profiles
- Identification of capacitive and inductive loads
- Performance monitoring for power factor correction

2. ***System-Level Analysis***: The overall system power factor is calculated using vector summation of individual load contributions, providing:

- Aggregate power factor monitoring
- System efficiency analysis
- Load balancing insights

C. Remote Control Implementation

The remote-control system enables intelligent load management through:

1. *Cloud-based Control*

- Arduino Cloud Integration: Secure remote switching capabilities
- Real-time Status Updates: Immediate feedback on control actions
- Scheduling Capabilities: Automated load management based on usage patterns

2. *Mobile Application Control*

- User Authentication: Secure access control for system management
- Intuitive Interface: User-friendly control and monitoring capabilities
- Real-time Notifications: Alerts for system status and anomalies

V. Results & Discussion

The developed smart energy meter which is capable of determining power factor and energy consumption, monitoring and controlling of individual loads is shown below in Fig. 3, and the internal circuitry is shown in Fig. 4. The system was tested with various loads including LED lights, electric bulbs, and inductive loads.

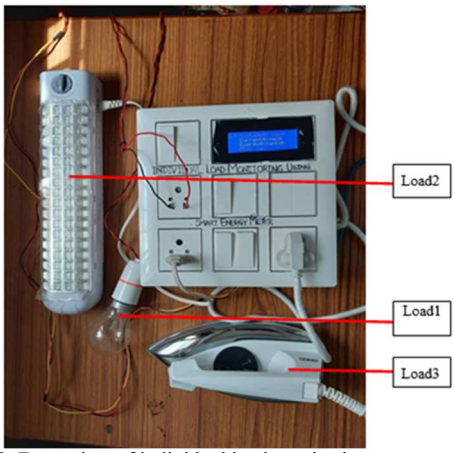

Figure 3: Front view of individual load monitoring smart energy meter

For calculating the voltage and current to used a multifunctional sensor known as PZEM004T sensor which is capable of sensing 80-260V voltage and 100A current. The PZEM004T energy monitor captures data from the loads connected to the terminals, including values of power (W), voltage (V), current (A), energy (kWh), frequency (Hz), and power factor.

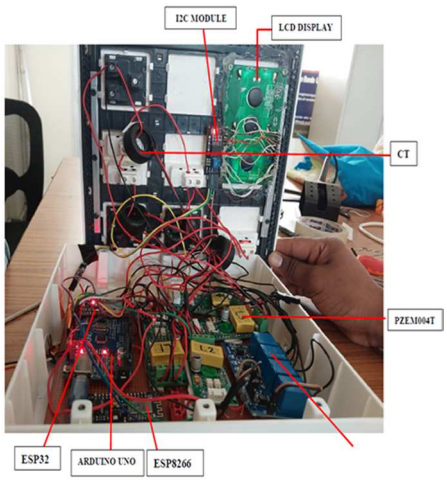

Figure 4: Internal circuitry of individual load monitoring using smart energy meter

A. Determination of Various Parameters of Different Loads

Authorized licensed use limited to: VTU Consortium. Downloaded on March 26,2026 at 15:28:08 UTC from IEEE Xplore. Restrictions apply.

The developed smart energy meter is used for three different loads (rechargeable led light ,15W electric bulb and choke coil) by using three PZEM004T sensors and a fourth sensor is determining the overall power factor and energy consumption. The voltage, current, energy consumption and power factor of a rechargeable led light, 15W electric bulb, a choke coil and overall power factor of these connected devices has been shown in Fig. 5 to Fig. 8.

Figure 5: Electrical parameters when load1 used as a rechargeable light

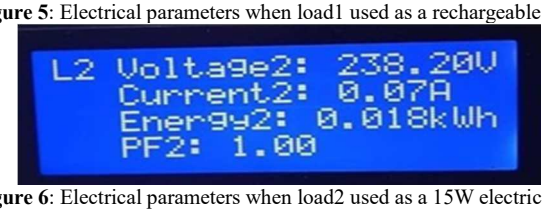

Figure 6: Electrical parameters when load2 used as a 15W electric bulb

Figure 7: Electrical parameters when load3 used as a choke coil

Figure 8: Electrical parameters when all three-load s connected

B. *Load Monitoring using ThingSpeak*

The system is able to monitor the loads remotely by using the "ThingSpeak" website. For this purpose, it is required to create a channel in website from which API key will be generated which will be helpful in uploading the code to ESP32 for load monitoring. Here, we have our API key as "MPD9QSQEY619ZUE2" as shown in Fig. 9. There is a feature of MATLAB visualization in ThingSpeak using which the status of a load can be remotely tracked in graphical form in real-time. The status of power factor, voltage and current of load viewed on this website under channel name "Individual Load Monitoring Using Smart Energy Meter" as shown in Fig. 10.

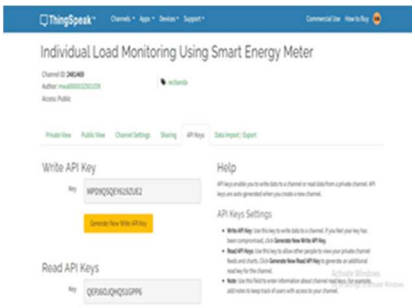

Figure 9: ThingSpeak interface API key

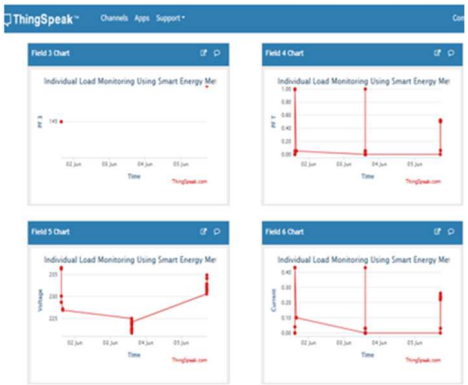

Figure 10: Status of power factor, voltage and current of load

C. *Load Controlling using Arduino cloud*

Here a Arduino cloud used for controlling various load from remote location. ESP8266 Wi-Fi module is used here to control the individual loads (i.e. ON or OFF state) of the proposed system, which is connected to individual load through a 4-channel relay. Using the internet connection ESP8266 module is connect to Arduino cloud. Fig. 11 shows the Arduino Cloud interface created for controlling the state of loads.

Here a 15W electric bulb was used as a load 1. Fig. 12 & Fig. 13 is used here to show the controlling the state of load 1 using Arduino Cloud dashboard. Fig. 12 shows that bulb did not glow when corresponding load 1 button of Arduino Cloud was in OFF condition. Fig. 13 shows when the bulb was turned ON through Arduino Cloud, the bulb glows.

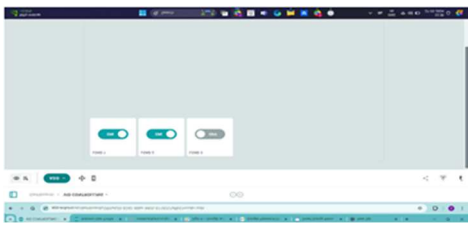

Figure 11: Load controlling interface using Arduino Cloud

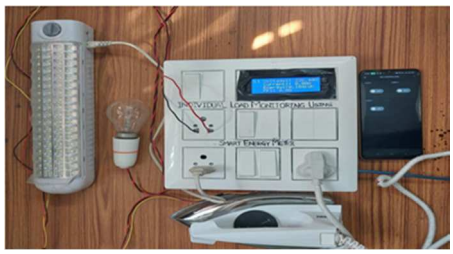

Figure 12: Bulb did not glow during OFF condition using Arduino Cloud

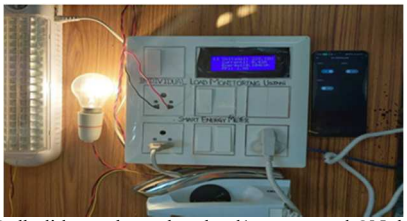

Figure 13: Bulb did not glow when load1 was turned ON through using Arduino Cloud

Authorized licensed use limited to: VTU Consortium. Downloaded on March 26,2026 at 15:28:08 UTC from IEEE Xplore. Restrictions apply.

D. Load Controlling using MIT Application

The real-time monitoring operation can be seen via a mobile application. MIT Application is used for the user login his/ her User ID and Password to access the detailed data of the loads. The interface of the MIT application is shown in Fig. 14. It includes seven fields: PF1, PF2, PF3, PFT, Voltage, Current, and Units. When load 1 was a mobile charger, load 2 was a 15 W bulb and load 3 was another mobile charger, the following data was obtained, which is shown in Fig. 15.

VI. Performance Analysis And Validation

A. System Accuracy Assessment

The implemented system demonstrates reliable performance across various load conditions. The PZEM004T sensors provide measurement accuracy within industry-standard tolerances, with voltage measurements accurate to ±1% and current measurements accurate to ±1% of full scale. Power factor measurements show consistent results across different load types, with particular accuracy for inductive loads where power factor correction is most critical.

B. Communication Performance

The multi-tier communication architecture demonstrates robust performance with minimal latency. UART communication between processing units maintains real-time data transfer rates, while WiFi connectivity to cloud platforms provides reliable remote access capabilities. The system maintains stable connectivity even under varying network conditions.

C. Control System Responsiveness

The relay-based control system demonstrates immediate response to remote commands, with switching times under 100ms. The cloud-based control interface provides reliable load management capabilities with real-time status updates.

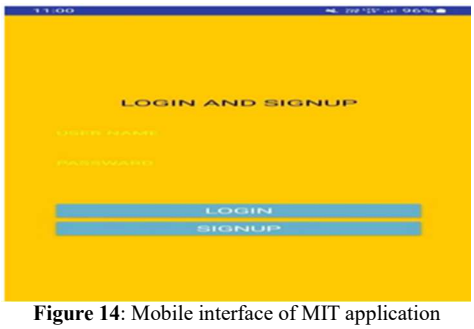

Figure 14: Mobile interface of MIT application

Figure 15: Real-time data of power factor of loads

VII. Conclusion

This research successfully demonstrates the development and validation of a comprehensive IoT-based smart energy monitoring system that addresses critical gaps in individual load monitoring and power factor management. The implemented system provides real-time monitoring of electrical parameters at the individual appliance level using multiple PZEM004T sensors integrated with Arduino UNO microcontroller and ESP modules. The system successfully integrates with ThingSpeak for data visualization and Arduino Cloud for remote control, providing users with comprehensive energy management capabilities accessible through both web and mobile interfaces.

The research contributes significantly to smart energy management through its innovative multi-sensor architecture for distributed sensing, device-level power factor monitoring and analysis, and cost-effective implementation using readily available components. Practical testing with various loads including LED lights, electric bulbs, and inductive loads validates the system's capability to accurately measure electrical parameters and provide meaningful insights into energy consumption patterns. The dual cloud platform integration demonstrates the feasibility of comprehensive energy management tools that enhance consumer awareness and enable targeted energy conservation efforts.

Future research opportunities include integration of machine learning algorithms for predictive energy management, development of robust security protocols, scalability enhancements for three-phase systems, and advanced features such as automated power factor correction and intelligent load scheduling. This research demonstrates that effective individual load monitoring can be achieved through accessible technologies while maintaining system reliability and user accessibility, contributing significantly to the advancement of smart energy management systems and providing a foundation for future IoT-based energy monitoring and control research.

References

Authorized licensed use limited to: VTU Consortium. Downloaded on March 26,2026 at 15:28:08 UTC from IEEE Xplore. Restrictions apply.

[1] P. Pawar, P. Vittal K, "Design and development of advanced smart energy management system integrated with IoT framework in smart grid environment," Journal of Energy Storage, vol. 25, 2019, https://doi.org/10.1016/j.est.2019.100846.

[2] Q. Sun et al., "A Comprehensive Review of Smart Energy Meters in Intelligent Energy Networks," in IEEE Internet of Things Journal, vol. 3, no. 4, pp. 464-479, Aug. 2016, doi: 10.1109/JIOT.2015.2512325.

[3] M. Milam and G. Kumar Venayagamoorthy, "Smart meter deployment: US initiatives," ISGT 2014, Washington, DC, USA, 2014, pp. 1-5, doi: 10.1109/ISGT.2014.6816507.

[4] P. Bharti, A. Singh, P. Singh, "Smart-UPS with a priority-based load management system," Journal of Physics: Conference Series, vol. 2007, 2021, DOI 10.1088/1742-6596/2007/1/012006.

[5] X. Fang, S. Misra, G. Xue and D. Yang, "Smart Grid — The New and Improved Power Grid: A Survey," in IEEE Communications Surveys & Tutorials, vol. 14, no. 4, pp. 944-980, Fourth Quarter 2012, doi: 10.1109/SURV.2011.101911.00087.

[6] A.E. Moore., "The history of the integrating electricity meter", Journal of the Institution of Electrical Engineers, vol. 77(468) , pp. 851-859, 1935, https://doi.org/10.1049/jiee-1.1935.0210.

[7] M. Karpagam, S. S. Sahana, S. Sivadharini and S. Soundhariyasri, "Smart energy meter and monitoring system using internet of things (IoT)", Proceedings of the International Conference on Intelligent Data Communication Technologies and Internet of Things (IDCIoT 2023), pp. 75-80, 2023, DOI: 10.1109/IDCIoT56793.2023.10053541.

[8] N. Mondal, M. Mohammed, H. Shake, H.A. Mahdi and C. Ravisankar, "Design of a smart energy meter," International Journal of Current Science, vol. 12(2), pp. 316-323, 2022.

[9] P. K. Sebastian and K. Deepa, "Internet of things based smart energy meter with fault detection feature and theft detection", 2022 International Conference on Electronics and Renewable Systems (ICEARS), pp. 494-500, 2022, DOI: 10.1109/ICEARS53579.2022.9751828.

[10] P. Selvam, G. Ramakrishnaprabu, K. Senthilkumar, R. Satish, R. Devarajan and P. Loganathan , "Intrusive energy management with advanced smart metering and monitoring using IoT", 2021 IEEE International Power and Renewable Energy Conference (IPRECON), pp. 359-364, 2021, DOI: 10.1109/IPRECON52453.2021.9640636.

[11] M. Orlando, A. Estebsari, E. Pons, M. Pau, S. Quer and M. Poncino, "A smart meter infrastructure for smart grid IoT applications", IEEE Internet of Things Journal, vol. 9(14), pp. 12529-12541, 2022, DOI: 10.1109/JIOT.2021.3137596.

[12] M.U. Saleem, M.R. Usman and M. Shakir, "Design, implementation, and deployment of an IoT based smart energy management system", IEEE Access, vol. 9, pp. 59649-59664, 2021, DOI: 10.1109/ACCESS.2021.3070960.

[13] R. Sheeba, N. Naufal, S.N. Beevi, A.R. Nair, S. Amal and A.S. Kumar, "Real-time monitoring of energy meters using cloud storage", 2021 IEEE International Power and Renewable Energy Conference (IPRECON), pp. 1- 5, 2021, DOI: 10.1109/IPRECON52453.2021.9640636.

[14] W.W.V. Souza, M.S. A. Ali, A.F.M. Borges, J.P.P. Oliveira, D.R.C. Silva and M.B. Nogueira, "Development of an energy meter based on IoT", 2020 IEEE International Workshop on Metrology for Industry 4.0 & IoT, 2020, pp. 241-244, DOI: 10.1109/MetroInd4.0IoT48571.2020.9138257.

[15] H.A. Hseiki, A.M. El-Hajj, Y.O. Ajra, F.A. Hija and A.M. Haidar, "A secure and resilient smart energy meter", IEEE Access, vol. 12, pp. 3114-3125, 2024, DOI: 10.1109/ACCESS.2023.3349091.

Authorized licensed use limited to: VTU Consortium. Downloaded on March 26,2026 at 15:28:08 UTC from IEEE Xplore. Restrictions apply.